6 A railway depot uses an electrical de-icing unit to prevent ice from forming on a track-switch mechanism. The unit contains a resistive heating element supplied by an ideal step-down transformer. A diagram of the setup is shown in Fig. 6.1 below.

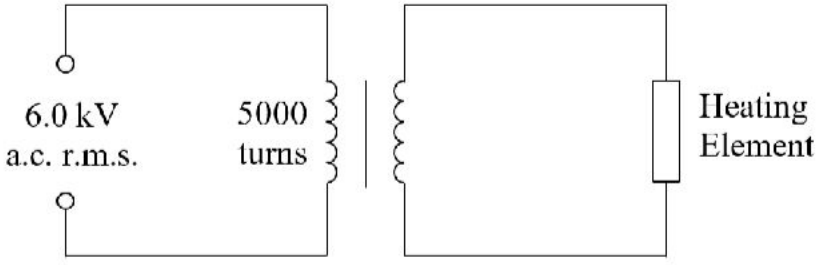


Fig. 6.1

The primary coil of the transformer is connected to a 6.0 kV, 50 Hz sinusoidal alternating supply. The potential difference across the secondary coil is 240 V r.m.s.. The primary coil has 5000 turns.

(a) Determine the number of turns on the secondary coil.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$N_s = 5000\left(\frac{240}{6000}\right)$$

$$= 200 \text{ turns} \qquad [1]$$

number of turns= ..... [1]

(b) The heating element behaves as a pure resistor and dissipates a mean power of 4.6 kW. The instantaneous current I in the heating element is represented by

$$I = I_o \sin \omega t .$$

Determine I_o and ω.

$$P_{\text{mean}} = V_{rms} I_{rms}$$

$$I_{rms} = \frac{4600}{240} = 19.17 \text{ A} \qquad [1]$$

$$I_{rms} = \frac{I_0}{\sqrt{2}}$$

$$I_0 = \sqrt{2} \times 19.17 = 27.1 \text{ A} \qquad [1]$$

$$\omega = 2\pi f = 2\pi(50) = 314 \text{ rad s}^{-1} \qquad [1]$$

I_o = ... A

ω = ... rad s^{-1} [3]

(c) Determine the r.m.s. current in the primary coil of the transformer.

$$V_p I_p = V_s I_s$$

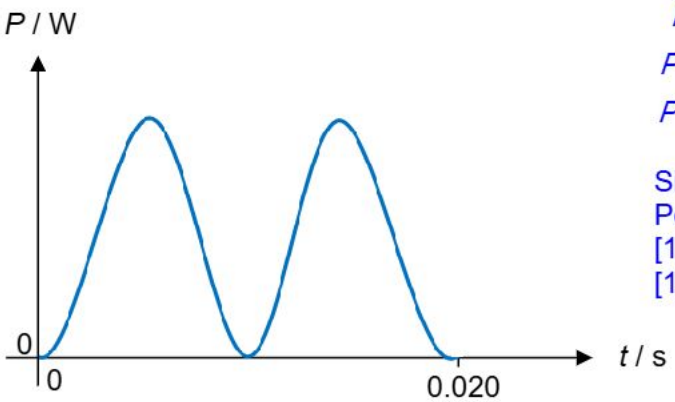
$$I_p = \frac{V_s I_s}{V_p}$$

$$= \frac{(240)(19.17)}{6000} \quad [1]$$

$$= 0.767 \text{ A} \quad [1]$$

current = ... A [2]

(d) On Fig. 6.2, sketch the variation with time t of the power P dissipated in the resistor from $t = 0$ to $t = 0.020$ s.



$I = I_0 \sin \omega t$

$P = I^2 R$

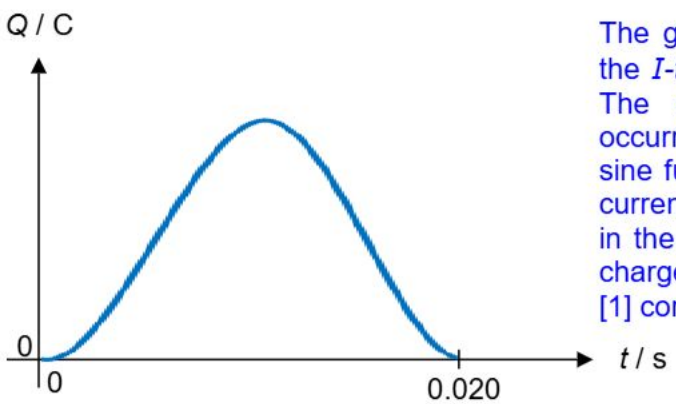
$P \propto \sin^2 \omega t$

Since the frequency is 50 Hz, the Period of the sine function is 0.02 s.
[1] for correct shape
[1] for correct period

Fig. 6.2

[2]

(e) On Fig. 6.3, sketch the variation of the net amount of charge Q which flowed past a point in the resistor from $t = 0$ to $t = 0.020$ s. (Numerical values for the charge are not expected).



The gradient of the Q-t graph gives the I-t graph, which is a sine graph. The Q-t graph has a maximum occurring at half of the period of the sine function. After half a period, the current reverses, so the charges flow in the opposite direction and the net charge flowed decreases until zero.
[1] correct shape

Fig. 6.3

[1]

[Total: 9]